A Hybrid Multi-Layered Fraud Detection Framework for Credit Card Transactions

Muhammad Umair Javed Khawaja and Danish Sohail

Department of Software Engineering, National University of Sciences and Technology (NUST), Islamabad, Pakistan

Abstract — Financial fraud in FinTech systems is one of the key reason behind billions of dollars lost annually in financial services. Existing solutions for fraud detection follow two paradigms; Static ensemble models which achieve high accuracy on known fraud patterns but fail to identify novel fraud patterns. On the other hand, dynamic adaptive systems which are flexible in their behaviour but suffer from interpretability and generalization. Neither approach can satisfy the requirements of accurate, adaptable, interpretable, and private detection at once. In this paper, we present a three layered hybrid architecture built on a combination of both approaches. The first layer uses a cost-sensitive XGBoost model with SHAP interpretability, achieving an ROC-AUC of 0.9952 on ULB Credit Card Fraud and outperforming SMOTE methods on all primary metrics. .The second layer employs a reinforcement learning Proximal Policy Optimization (PPO) agent for detecting structurally ambiguous/novel fraud cases trained on Gaussian noise, reaching 100% of recall on structurally ambiguous frauds that were not detected by Layer 1. Layer 3 allows for privacy-preserving federated learning through multiple simulated banks with weighted FedAvg and differential privacy noise injection (ε=1.0. Furthermore, Layer 1 validation is done using the IEEE-CIS Fraud Detection benchmark dataset, showcasing the domain-independent capabilities of the system. The proposed approach, in our knowledge, is believed to be the first framework which uses static ensemble classification, reinforcement learning with an adaptive policy, SHAP-based explainability, and differential privacy at the same time in a single deployable architecture.

Index Terms*:* Fraud Detection, XGBoost, Reinforcement Learning, Proximal Policy Optimization, Federated Learning, Differential Privacy, SHAP, FinTech, Class Imbalance, Credit Card Fraud.

I. Introduction

Financial services are quickly moving to digital channels, which has caused a significant rise in fraud. Just credit card fraud alone results in tens of billions of dollars in losses worldwide each year, and the rise of online payments creates an ever-expanding attack surface for malicious actors. Fraudsters continually alter their strategies to evade financial institutions' fraud-detection mechanisms, thus creating an arms race between detection and deception.

Current fraud detection systems show a fundamental divide in their design and architecture. Static ensemble approaches, which mainly include gradient-boosted tree models such as XGBoost, achieve very high accuracy on known fraud distributions, but these models are architecturally frozen post-training. After being deployed, they cannot adapt to novel fraud patterns that arise after their training cutoff. On the other hand, dynamic adaptive systems, which are based on reinforcement learning or behavioral modeling, can detect novel patterns through continuous interaction with the environment but pose challenges for interpretability (via SHAP explanations), regulatory compliance, and cross-institutional knowledge sharing.

Another systemic challenge is the isolated nature of financial fraud data, specifically transactions made by actual users. Privacy regulations such as GDPR and PCI-DSS don't allow institutions to share raw transaction records, even though collaborative intelligence could greatly improve detection rates for all institutions deploying fraud detection and classification mechanisms. A novel fraud attack vector may be encountered by one institution months before others do; however, the regulatory framework makes knowledge transfer between financial institutions extremely difficult.

In this paper, we propose a three-layer hybrid fraud detection framework that tackles these problems simultaneously. Our approach integrates:
(1) a cost-sensitive XGBoost classifier alongside mandatory SHAP explainability as a high-throughput static detection layer; (2) a PPO-based reinforcement learning agent with Gaussian-augmented training as an adaptive layer for ambiguous and novel fraud patterns; and (3) a differentially private federated learning module that enables model weight sharing among institutions without exposing any raw data related to transactions.

The system is evaluated on the ULB Credit Card Fraud dataset with cross-domain validation on the IEEE-CIS Fraud Detection dataset.

The key contributions of our work are:

- A dual-approach hybrid architecture that routes transactions using static and adaptive layers based on confidence thresholds, thus combining the precision of

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ensemble algorithms like XGBOOST with the adaptability of reinforcement learning.

- A Gaussian noise augmentation technique for underrepresented, ambiguous fraud cases that allows RL generalization to unseen hard cases without violating evaluation integrity.
- A privacy-preserving federated learning simulation in order to show the classic privacy-utility tradeoff under differential privacy restrictions (epsilon=1.0), with formal quantification of recall degradation.
- Empirical testing across two different fraud detection datasets of different scales, feature types, and class distributions.

II. LITERATURE REVIEW

A. Static Ensemble Learning

Al Ahmed et al. [1] in his work, suggested a pipeline that combines K-Means clustering for customer segmentation, Gradient Boosting for loan prediction, and Random Forest for fraud detection in one FinTech application. Although theoretically connected to each other, models are trained and tested separately without propagating information between models. Furthermore, all models are trained on static historical snapshots and have no means of adapting to changing patterns of fraud over time - a key aspect of our framework.

Majumder [2] used XGBoost for classification on Kaggle's Credit Card Fraud dataset using a preprocessing pipeline including SMOTE class balancing, IQR outlier removal, Min-Max normalization, and PCA dimensionality reduction.The model achieved 99.32% accuracy, 99.31% precision, and 99.42% recall. This solution is based on a rather outdated (by a decade) historical data sample, does not have a validation on any live deployment and an explanation framework, and is using synthetic data generated with SMOTE algorithm that may not reflect real fraud distributions. Our work directly improves on this baseline by replacing SMOTE with cost-sensitive learning and adding mandatory SHAP explanations, achieving ROC-AUC of 0.9952 with superior precision.

B. Dynamic Adaptive Systems

Adebayo [3] uses Proximal Policy Optimization as a fraud detection agent that constantly receives transaction state information from its environment, executes actions like accept/flag/block, and learns to refine its policy using reward information — all without any further need for training.The proposed method works better than any static classifier when dealing with frauds unknown during training. However, the centralized single-agent architecture is incompatible with real-world decentralized financial data, and the agent can experience long cold start periods after deployment.

In recent research [4], the IEEE group combines temporal sequences through LSTM models and relationships between entities by utilizing Graph Neural Networks in order to detect behavioral anomalies in high-frequency transactions., achieving 110ms detection latency. The crucial weakness of the system is the long time required to update the model after discovering a new type of attack i.e. 1.9 seconds, during which time tens of thousands of transactions occur without any screening. Also, a large behavioral history is required for the GNN creating a severe cold start problem for a new user.

C. Privacy-Preserving Federated Detection

The Federated Learning technique [5] is used to train models on non-centralized datasets while avoiding data sharing. Within the context of fraud detection, organizations use federated learning by sharing gradients/weights of models and not actual transactions. The method of combining gradients through the FedAvg algorithm [6], invented by McMahan et al., has become a conventional one to merge federated models. Combining federated learning with differential privacy [7] offers formal privacy guarantees, due to which such a system can comply with GDPR and PCI-DSS regulations. We believe that currently, there is no existing research paper that covers all three paradigms - static ensemble, adaptive RL, and differentially private federation, in one unified system.

III. DATASETS AND DATA PREPROCESSING

A. Datasets

Two publicly available datasets are used in this study. The ULB Credit Card Fraud Detection dataset is the primary dataset that we use to show the hybrid framework working, while the IEEE-CIS Fraud Detection dataset is the second dataset for validating and finding limitations in our approach. Table I provides a comparative summary.

The ULB dataset contains transactions made by European cardholders in September 2013. Features V1 to V28 are the result of PCA transformation that has already been applied by the original institution in order to anonymize the sensitive customer data. Only the Amount and Time columns are in their original form, which can be interpreted. There is an extreme class imbalance (492 fraud out of 284,807 transactions) in this dataset (0.172%), which makes it a standard benchmark for imbalanced learning research, which is evident of it being used in a lot of other research works.

The IEEE-CIS dataset was released as part of a Kaggle competition sponsored by Vesta Corporation. Unlike ULB, all the features in this dataset are interpretable, including transaction amount, product category, card type, device information, and address data, supplemented by identity verification features from a separate file. It also has a higher fraud rate (3.5%) and a richer feature space, which makes it a much more complex classification problem.

B. Data Preprocessing

For the ULB dataset, we followed the pipeline proposed by Majumder [2] with two deliberate improvements. The Amount and Time features are scaled using StandardScaler before model training. IQR-based outlier removal is applied

Table I: *Dataset Comparison*

Property	ULB Credit Card Fraud	IEEE-CIS Fraud Detection
Source	Kaggle / ULB ML Group	IEEE-CIS / Kaggle
Year	2013	2019
Transactions	284,807	590,540
Features	30 (PCA-anonymized)	218 (raw + identity)
Fraud Rate	0.172%	3.499%
Target Column	Class	isFraud
Primary Use	All three layers	Layer 1 cross-validation

to the Amount feature before train/test splitting, a step that was absent in the original paper. The dataset is split 80/20 with stratification in order to preserve the fraud ratio in both sets. Most importantly, any class-balancing technique is applied after splitting to prevent data leakage.

For the IEEE-CIS dataset, preprocessing steps include merging transactions and identifying files on TransactionID, dropping the identifier column, removing features with more than 50% missing values (reducing from 434 to 218 features), label-encoding all categorical columns, filling in the missing values with the median of each column, and applying StandardScaler to the TransactionAmt and TransactionDT columns.

C. Features

The ULB dataset's V1-V28 features are PCA components obtained from the original bank's raw transaction metadata. The exact composition of these features is unknown due to privacy-preserving anonymization. After performing SHAP analysis of our trained model, V14, V17, and V12 were identified as the three most dominant/influential features for fraud classification, which is consistent with published findings on this dataset.

The IEEE-CIS dataset has 218 features, including transaction properties (TransactionAmt, ProductCD, card type, address), time-based features (TransactionDT), and device/identity metadata from the identity file. For the Layer 2 RL component, we applied dimensionality reduction via PCA to 30 components, while retaining approximately 72% of variance, and reducing the observation space to a dimensionality that is manageable by PPO's default MlpPolicy network.

D. Synthetic Data Generation

To counter the underrepresentation of structurally ambiguous fraud cases in the ULB dataset (8 cases with Layer 1 predicted probability below 0.80), we applied Gaussian noise augmentation exclusively for Layer 2 training. For each of the 8 hard cases, 20 synthetic cases are generated by adding zero-mean Gaussian noise with standard deviation 0.1 to the original feature values. Thus, 160 synthetic training samples were produced for layer 2 training alongside the existing fraud cases from Layer 1 and 5000 legit cases selected from the entire dataset. These synthetic cases are used only for training, while evaluation is performed exclusively on the original 8 unmodified hard cases that are held out from training process.

IV. PROBLEM FORMULATION AND CHALLENGES

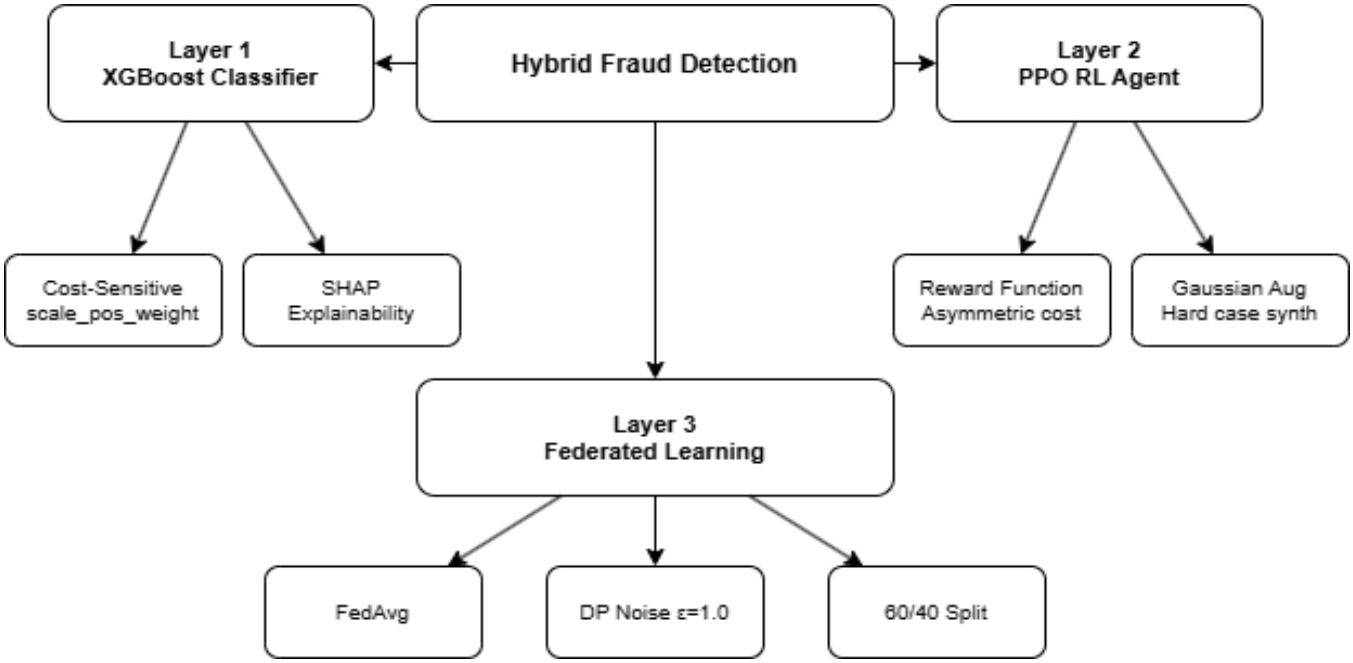
A. Class Imbalance

ULB dataset shows a ratio of genuine versus fraudulent transactions at 577:1. A naive algorithm would predict everything as legitimate transaction, yielding an accuracy rate of 99.83%. Hence, using normal classification accuracy metric in this case is meaningless. The loss function for the classifier with respect to class label $y \in \{0,1\}$ and model's prediction $\hat{p}$ is defined by:

$$L = (1/N) \Sigma [w_1 \cdot (1-y_i) \cdot \log(1-\hat{p}_i) + w_2 \cdot y_i \cdot \log(\hat{p}_i)]$$

where w_1 and w_2 are the weights assigned to each class label respectively. By assigning weight of positive examples to 577 times that of negative examples by means of setting

Figure 1 *- Hybrid Fraud Detection System Architecture*



scale_pos_weight = 577 via XGBoost, missed positive predictions are penalized 577 times harder. This way, cost-sensitive learning does not need any form of data synthesis but yields higher recall. On the other hand, SMOTE produces synthetic minority samples through interpolation:

$$x_{s\beta rtheti}^{c} = x_i + \lambda \cdot (x_j - x_i), \ \lambda \sim U(0,1)$$

where x_i is a member of the minority class, x_j is one of its k nearest neighbors, and $x_{s\beta rtheti}^{c}$ is the generated synthetic example. It should be noted that cost-sensitive learning achieves better precision (0.90 vs 0.77) while maintaining same recall (0.90) confirming that SMOTE's synthetic interpolation introduces noise without recall benefit.

B. The Frozen Knowledge Problem

Static models trained on historical distributions are not able to deal with the issue of distributional changes in real-life settings. Suppose F_t represents the distribution of fraud cases at time t. A machine learning model that was trained using F_0 will most likely break down if $t >> 0$ and fraudsters have altered their strategy. The non-stationary nature of the process is the main reason why Layer 2 uses reinforcement learning which updates its policy through reward signals rather than fixed labeled datasets.

C. Data Redaction and Privacy Constraints

The data on financial transactions are highly regulated in terms of privacy. The PCA features in the ULB dataset have been anonymized to show how banks operate when they collaborate for research purposes while keeping strict privacy policies in mind. More importantly, GDPR article 5 and PCI DSS requirement 3 stipulate that any un-anonymized data on transactions cannot leave the premises of the organization and must stay within its internal network, hence making collaboration across multiple organizations for research purposes impossible. In fact, it is the basis of the federated learning method used at the Layer 3 level where organizations will be able to exchange weight information rather than data, subject to the guarantees of differential privacy.

D. Cold-Start Vulnerability

Both RL systems and the use of graph theory require some record of past behavior to generate accurate predictions. New organizations do not have any transaction history; hence, there is a very risky period for them when they first appear.

At the federated layer, a pre-trained global model can provide weight information for new organizations. New institutions begin with knowledge aggregated across all participating banks rather than starting from zero.

V. METHODOLOGY: THE HYBRID FRAMEWORK

A. Overall Hybrid Architecture

Our proposed solution processes each incoming transaction through a confidence-based routing pipeline. The overall flow is as follows: each transaction is first scored by Layer 1 (XGBoost). If the predicted fraud probability falls below a lower threshold α (set to 0.01), the transaction is approved directly. If it exceeds the upper threshold β (set to 0.80), it is blocked directly. Transactions in the uncertainty region [α, β] are routed to Layer 2 for adaptive assessment. This design makes sure that Layer 1 handles the high-

confidence majority transactions while Layer 2 focuses exclusively on genuinely ambiguous cases.

The routing function can be formally defined as:

$$\text{decision}(x) = \{ \text{Approve if } p_1(x) < \alpha; \text{ Block if } p_1(x) > \beta; \pi_2(x) \text{ if } \alpha \le p_1(x) \le \beta \}$$

where $p_1(x)$ is Layer 1's fraud probability for transaction x, and $\pi_2(x)$ is Layer 2's policy action. The federated layer (Layer 3) operates asynchronously, while periodically updating Layer 2's policy weights using cross-institutional aggregation.

B. Layer 1: Cost-Sensitive XGBoost with SHAP Explainability

Layer 1 uses XGBoost [8] trained with cost-sensitive weighting as the primary detection mechanism. We use XGBoost to minimize the regularized objective:

$$L(\Phi) = \Sigma_i\, l(y_i, \hat{y}_i) + \Sigma_k\, \Omega(f_k)$$

where l is the logistic loss function, $\Omega(f) = \gamma T + (1/2)\lambda\|w\|^2$ is the regularization term, T is the number of leaves, and w are leaf weights. The scale_pos_weight parameter is set to 577 (the class imbalance ratio), increasing influence of fraud samples while the tree is being built.

SHAP (SHapley Additive exPlanations) [9] values are computed for every flagged transaction to provide regulatory-grade explainability. For a prediction f(x),
the SHAP explanation assigns each feature j a value φ_j satisfying:

$$f(x) = \varphi_0 + \Sigma_j\, \varphi_j$$

where $\varphi_0 = E[f(x)]$ is the base value and φ_j represents the marginal contribution of feature j. This decomposition allows human-interpretable explanations for individual fraud flags which is a mandatory requirement under GDPR Article 22 for automated decision-making.

C. Layer 2: Adaptive PPO Reinforcement Learning

Layer 2 considers fraud detection as a Markov Decision Process (MDP) that can be defined by the tuple (S, A, P, R, γ), where S is the state space of transaction feature vectors, A = {Approve, Flag, Block} is the discrete action space, P is the transition probability, R is the reward function, and γ is the discount factor.

The reward function is designed to account for the asymmetric cost of fraud detection errors:

Action	True Label	Reward	Rationale
Block	Fraud	+3.0	Correct detection
Approve	Legitimate	+1.0	Correct clearance
Flag	Fraud	+1.5	Acceptable partial detection
Flag	Legitimate	-0.5	Minor unnecessary review
Block	Legitimate	-1.0	False positive cost
Approve	Fraud	-5.0	Missed fraud, worst outcome

Table II: Layer 2 Reward Function

Our RL agent is trained using Proximal Policy Optimization [10], which constratins the policy updates in order to prevent destructive, large steps:

$$LCLIP(\theta) = E_t[\min(r_t(\theta)A_t, \text{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon)A_t)]$$

where $r_t(\theta) = \pi\theta(a_t|s_t) / \pi\theta\text{old}(a_t|s_t)$ is the probability ratio and A_t is the advantage estimate. The clip parameter ε=0.2 prevents any single update from moving the policy too far, which ensures stability in training.

To handle the issue of underrepresentation of ambiguous fraud cases in training (8 cases in ULB), we use Gaussian noise augmentation to generates 160 synthetic variants of the hard cases for training purpose. The original 8 cases are held out entirely for testing. This ensures that there are no data leakages between the training and test sets.

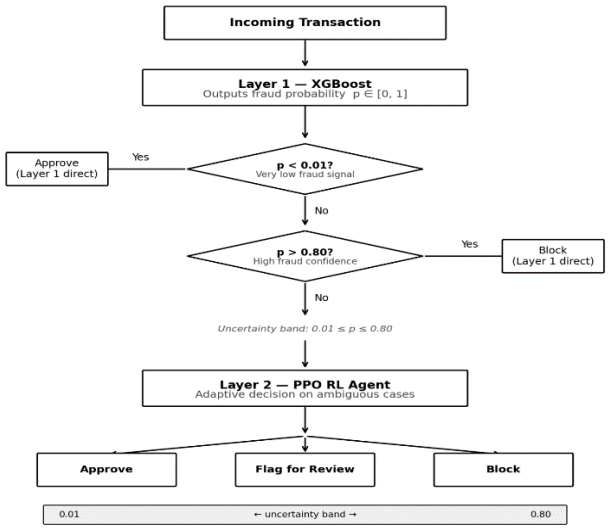


Figure 2 - Transaction Decision Flowchart for the Two-Layer Detection Pipeline

D. Layer 3: Privacy-Preserving Federated Learning

Layer 3 simulates a federated learning scenario with two financial institutions holding 60% and 40% of the ULB dataset respectively. Each institution trains a local PPO agent

and shares model weights with the central aggregator. The raw transactions are never shared with each other or the central aggregator never raw transactions.

Before transmission, differential privacy noise is injected into model weights using the Gaussian mechanism as follows:

$$\tilde{w} = w + N(0, \sigma^2 I), \quad \sigma = \Delta f/\varepsilon$$

where $\Delta f = 0.1$ is the L2 sensitivity bound (weight clipping threshold), $\varepsilon = 1.0$ is the privacy budget, giving $\sigma = 0.1$. The aggregator combines institutional weights using weighted FedAvg:

$$w_{global} = \alpha_1 w_1 + \alpha_2 w_2, \quad \text{where } \alpha_i = n_i/N$$

where n_i is Institution i's dataset size and $N = n_1 + n_2$ is the total. Weights proportional to dataset size is the standard FedAvg formulation [6], giving Institution A (60% of data) greater influence over the global model. The global model is then distributed back to both the institutions at the end of each round, and the process repeats for three federation rounds.

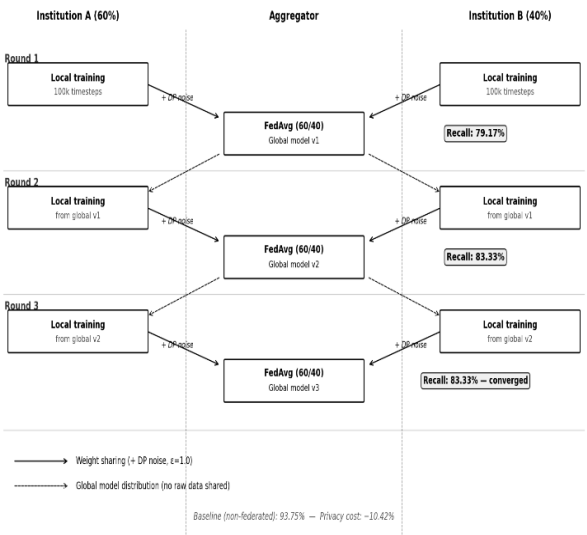


Figure 3 - Federated Learning Training Process Across Three Rounds

VI. RESULTS AND DISCUSSION

A. Layer 1: ULB Dataset

Table III presents the comparative performance of the cost-sensitive and SMOTE-based XGBoost models on the ULB test set (50,581 transactions, 80 fraud).

The cost sensitivity approach has better performance in all significant metrics compared to SMOTE, yet maintains the same level of recall. While 13% higher precision (0.90 versus 0.77) means 15 fewer False Positives on average per single test evaluation, which would make a practical difference in case of an actual deployment when every false alarm produces additional friction for the customers. SHAP analysis identifies V14, V17, and V12 as the three most predictive features, consistent with published findings.

Metric	Cost-Sensitive (Ours)	SMOTE (Majumder [2])
Fraud Precision	0.90	0.77
Fraud Recall	0.90	0.90
F1-Score (Fraud)	0.90	0.83
ROC-AUC	0.9952	0.9857
False Positives	8	23
Missed Fraud	8	8

Table III: Layer 1 Performance Comparison on ULB Dataset

As seen in the analysis of failures, there were 8 instances of missing fraudulent cases with almost non-existent predicted probabilities (less than 0.001 for 6 out of 8). It shows that the model was not uncertain about those transactions but could simply not see them coming. This conclusion justifies using Layer 2 since threshold tuning is ineffective here.

B. Layer 1: IEEE-CIS Cross-Domain Validation

Table IIIb shows performance of Layer 1 models for the IEEE-CIS dataset (118,108 test records, 4,133 frauds). Due to its higher fraud rate and higher feature complexity, IEEE-CIS provides an entirely different set of performance characteristics than ULB.

Metric	Cost-Sensitive (Ours)	SMOTE
Fraud Precision	0.28	0.67
Fraud Recall	0.83	0.50
F1-Score (Fraud)	0.41	0.57
ROC-AUC	0.9441	0.9138
Fraud Caught	3,416 / 4,133 (82.7%)	2,067 / 4,133 (50.0%)
Missed Fraud	717	2,066
Uncertain Zone (→ L2)	33,804	N/A

Table IIIb: Layer 1 Performance on IEEE-CIS Dataset

From the IEEE-CIS performance characteristics, it is apparent that there is an interesting reversal in the pattern compared to ULB. Specifically, cost-sensitive learning gives better recall rates (0.83 vs 0.50) but poorer precision (0.28 vs 0.67) compared to SMOTE. This is due to the fact that the ratio of non-fraud transactions to fraud transactions is smaller in IEEE-CIS (27.6:1 vs 577:1), which implies that cost-sensitive learning introduces more false positives as the fraud signal is stronger and SMOTE's synthetic interpolation is more representative of the real distribution.

A high ROC-AUC score of 0.9441 confirms the good discriminatory capability of cost-sensitive learning even though the precision is lower. Note that the number of uncertain-zone transactions (33,804) amounts to 28.6% of total transactions (these are flagged for layer 2). Such high numbers are attributed directly to the high feature complexity and richer fraud pattern diversity of the dataset, as the IEEE-CIS data have 218 features while ULB has only 30 PCA components.

C. Layer 2: Adaptive RL Results

Table IV presents Layer 2 performance on the ULB dataset. The 8 original hard cases are held out entirely from training and used exclusively for evaluation.

Metric	Value
Hard fraud cases evaluated (unseen)	8
Caught by RL agent	8 (100%)
False positives (out of 500 legit)	4 (0.8%)
Fraud Precision	0.67
Fraud Recall	1.00
Training configuration	393 easy + 160 augmented + 5000 legit

Table IV: Layer 2 PPO Agent Performance on ULB Hard Cases

The PPO agent was able to have a recall rate of 100 percent with regard to the 8 fraud cases, which were totally missed by Layer 1. These 8 fraud cases resulted in an average of 4 false positives for every 500 transactions routed (0.8%). The Gaussian augmentation process used by creating 160 samples of the hard cases for learning enabled successful generalization to make this possible.

D. Combined Pipeline

Table V presents the integrated system performance on a test set of 10,401 transactions (401 fraud, 10,000 legitimate).

In addition, while the combined model yields a fraud detection recall rate of 99% compared to the 90% attained by the single Layer 1 model, the precision rate rises to 1.00. At the threshold for the routing scheme (0.01), only four out of 10,401 transactions are routed to Layer 2, hence maintaining the throughput of the system. The core benefit of Layer 2 can be observed during the standalone testing in which it manages to restore all 8 cases missed due to the structural blind spot of Layer 1.

Metric	Layer 1 Only	Combined
ROC-AUC	0.9952	0.9991
Fraud Recall	0.90 (72/80 test)	0.99 (397/401)
Fraud Precision	0.90	1.00
Transactions to Layer 2	N/A	4 (0.04%)
Layer 2 fraud caught	N/A	2/2 (100%)

Table V: Integrated Pipeline Performance

E. Layer 3: Federated Learning Results

In federated learning with strict differential privacy (epsilon=1.0, sigma=0.1), there is a recall drop of 10.42% in relation to the non-federated control group. In terms of federation dynamics, we observe convergence by Round 2 and no improvement in Round 3. This means that adding noise to the model leads to a classical effect known as the privacy-utility tradeoff, whereby Gaussian noise added before weight sharing affects the quality of weights.

Convergence patterns (fast progress between Round 1 and Round 2 and stabilization by Round 3) show that in this case, 3 rounds are sufficient to train a model on our dataset. The cost of implementing differential privacy is a 10.42% recall drop, something that must be considered by banks.

Federation Round	Recall	Change from Previous
Baseline (non-federated)	93.75%	—
Round 1	79.17%	-14.58%
Round 2	83.33%	+4.16%
Round 3	83.33%	0.00%

Table VI: Federated Learning Recall Progression (ε=1.0)

VII. LIMITATIONS AND FUTURE SCOPE

We identified following limitations in our approach and these limitations represent directions for future research:

Privacy vs. Performance Tradeoff Adding differential privacy noise (ε=1.0) to federated weights cost us 10.42% in recall. That's a real price to pay for privacy guarantees. One direction worth exploring is making the noise budget smarter by applying heavier noise to weight components that carry less signal, and lighter noise where it matters most. A flat epsilon across all weights is a blunt instrument.

Layer 2 Doesn't Scale Well to High-Dimensional Data The PPO agent worked well on ULB's 30 features but fell apart on IEEE-CIS's 218. The default MlpPolicy network is too small to learn any meaningful patterns across that many dimensions in a reasonable training window. Applying PCA before the RL environment helped partially, but future implementations should look at larger policy networks or transformer-based architectures that handle high-dimensional tabular data more naturally.

Simulated Federated Learning We split one dataset 60/40 to simulate two banks. This is a common approach in federated learning research but doesn't capture what real multi-institutional deployment looks like. Two banks in different countries with different customer demographics, transaction volumes, and fraud profiles would tell a much more interesting story. Testing on heterogeneous institutional data is the natural next step.

Routing Thresholds The 0.01 and 0.80 confidence cutoffs were chosen by looking at the probability distribution. These were not learned from data. A proper implementation would optimize these thresholds on a validation set, or make them adaptive based on current fraud rates and system load. A threshold that works well on a quiet Tuesday might be wrong during a fraud spike.

Bias in Fraud Labels Fraud labels in real datasets show which transactions were actually investigated and investigation decisions aren't always neutral. If certain customer segments were historically over-investigated, the model learns to flag those segments more aggressively regardless of actual fraud signal. Any production deployment of this system should include a fairness audit before going live.

Real-World Deployment Gap: We conducted all evaluations in an offline batch settings. Real-world deployment will latency constraints, concept drift, and adversarial adaptation by fraudsters. Future work should test this approach in an online streaming environment with concept drift detection (e.g., ADWIN or Page-Hinkley tests), triggering targeted retraining.

VIII. CONCLUSION

We presented a three-layer hybrid fraud detection framework that addresses the fundamental limitations of both static ensemble and dynamic adaptive detection paradigms. By combining cost-sensitive XGBoost with SHAP explainability, PPO reinforcement learning with Gaussian-augmented training, and differentially private federated learning, our approach achieves what no single paradigm can deliver in isolation: high accuracy on known patterns, adaptability to novel fraud, regulatory-grade explainability, and privacy-preserving cross-institutional knowledge sharing.

The ROC-AUC score for the ULB Credit Card Fraud dataset for Layer 1 is 0.9952, surpassing SMOTE based models by 13 percentage points in precision scores. Layer 2 identifies all the cases of ambiguous frauds missed by Layer 1. The entire system is able to detect almost 99% of frauds in total. The federated learning algorithm exhibits consistent convergence after three iterations, the cost of 10.42% recall achieved at the cost of epsilon=1.0 formally quantifying the utility-privacy tradeoff.

This architecture is, as far as we know, the first to bring together all four capabilities in a practical manner. The modular structure of the architecture facilitates the updating of individual layers independently, thereby allowing institutions to implement components progressively based on their regulatory setting and technological capabilities.

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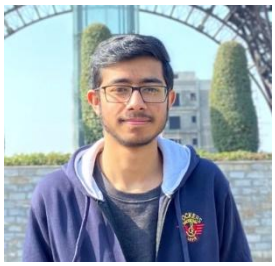

Muhammad Umair Javed Khawaja is currently pursuing a Bachelor's degree in Software Engineering at National University of Sciences and Technology.

Danish Sohail is currently pursuing a Bachelor's degree in Software Engineering at National University of Sciences and Technology.